

PAPER_1967 — The β_i Four-Channel Buoyancy Decomposition is a Coupled Ub-Vector, Not a Lattice of Independent Phenomenological Bounds — Full β_i Infrastructure Buried in PAPER_1167 UPDATE (Master Lagrangian) + PAPER_1168 (Falsifiable Prediction P4 $\pm 0.5\%$) + PAPER_1169 (10-System Numerical Confrontation) — PAPER_1966 Draft 3 Retraction of Implied ' $\beta_2 = 0.45$ Uncatalogued' Claim

Daniel T. Murphy

PAPER_1967 — β_i Four-Channel Decomposition Infrastructure

Abstract

Following a course-update from the project owner (" β_i has a full range derivation. It is most likely buried in the whitepapers"), a deep search of the whitepaper corpus surfaced a complete β_i falsifiable-prediction infrastructure that was overlooked during PAPER_1966's Draft 2 / Draft 3 prior-art check. The infrastructure spans four papers:

- **PAPER_1165 (session 252, 2026-05-10)** — the triangular closure $\beta_i = 3(5-i)/20$ giving $\{0.6, 0.45, 0.3, 0.15\}$; already acknowledged in PAPER_1966 Draft 2.
- **PAPER_1167 UPDATE (Master Lagrangian 6-Term Form)** — wires the β_i ladder into the closed 6-term L_{UQFF} as $\Sigma_i \beta_i U_g U_b$, with variational stationarity $\delta S / \delta \phi = 0$ enforcing $F_U = 1$ globally.
- **PAPER_1168 (Falsifiable Predictions of the Closed Lagrangian)** — establishes **Prediction P4**: $\{\beta_1, \beta_2, \beta_3, \beta_4\} = \{0.603, 0.450, 0.300, 0.150\} \pm 0.5\%$ (the $SO(5)^2$ -correction band = $1/200$). Any UQFF astrophysical fit with β_i deviating by $>0.5\%$ falsifies G2 closure and the master synthesis PAPER_1167.
- **PAPER_1169 (Numerical Confrontation P1-P5 with Archival Data)** — performs the actual re-fit test on **10 UQFF-modeled systems** (Sgr A*, M87 SMBH, SGR1745-29, AT2019qiz, ASKAP J1832-0911, Helix Nebula, Pillars of Creation, Westerlund 2, NGC 3596, Tapestry SFR). **All 10 recover β_i within $\pm 0.5\%$.** Best-fit β_2 values across the 10 systems: (0.449, 0.451, 0.450, 0.448, 0.451, 0.450, 0.450, 0.450, 0.451, 0.449). Verdict: [OK] P4 confirmed.

Consequences for PAPER_1966:

1. **Retraction of Draft 3's implied " $\beta_2 = 0.45$ remains uncatalogued" claim.** PAPER_1169 already validated $\beta_2 = 0.45$ across 10 UQFF systems eight months before PAPER_1966 was authored.
2. **Refinement of PAPER_1966's structural-identity hypothesis.** The β_i are NOT four independent phenomenological bounds requiring separate anchor identification. They are the **four-channel weights of a single coupled Ub decomposition**, tested falsifiably as a single unit. What Draft 3 called "3-of-4 concordance" was actually a "channel-projection observation": specific observables (starburst M_{sf} , PAPER_1953 0.3 factor) happen to sit near the dominant β_4 or β_3 channels of the decomposition. This is a fundamentally different claim from " β_i lattice of independent bounds."

Positioning (honest scope after course-update): This paper does not introduce any new derivation. It (i) surfaces the pre-existing β_i infrastructure that PAPER_1966 overlooked, (ii) retracts and corrects PAPER_1966 Draft 3's framing, and (iii) recontextualizes the $M_{sf} \equiv \beta_4$ and PAPER_1953 $\leftrightarrow \beta_3$ identifications as channel-projection observations within the falsifiable 4-channel model.

1. Background — What PAPER_1966 Said

PAPER_1966 (2026-07-09) documented that $M_{sf} = 0.15 = 3/(2 \cdot S0_5)$ for the starburst mass-fraction bound established in PAPER_774/PAPER_784. Draft 2 acknowledged that PAPER_1165 already contained the arithmetic identity $\beta_4 = 3/20 = 0.15$. Draft 3 hypothesized a "structural-identity" between the β_i ladder and separately-documented UQFF phenomenological bounds, verifying:

- $\beta_1 = 0.6 \leftrightarrow$ canonical $\beta_i = 0.6029$ (PAPER_1203) — confirmed within 0.5%
- $\beta_2 = 0.45$ — "remains uncatalogued"
- $\beta_3 = 0.3 \leftrightarrow$ PAPER_1953 0.3 factor cross-regime universality — confirmed matching value
- $\beta_4 = 0.15 \leftrightarrow M_{sf}$ starburst bound (PAPER_774/PAPER_784) — this paper's finding

Draft 3 concluded: "3-of-4 concordance strengthens the structural-identity hypothesis."

The Draft 3 prior-art search examined the β_i triangular closure paper (PAPER_1165) but did not surface PAPER_1167 UPDATE, PAPER_1168, or PAPER_1169. This was a gap.

2. The Course-Update Discovery

Following the project owner's course-update (" β_i has a full range derivation. It is most likely buried in the whitepapers"), a targeted search on the pattern " $\beta_1 \beta_2 \beta_3 \beta_4$ " surfaced three additional papers not covered by PAPER_1966's search:

2.1 PAPER_1167 UPDATE — β_i in the Master Lagrangian

PAPER_1167 UPDATE ("Master Lagrangian 6-Term Form", session June 2026) integrates β_i into the closed 6-term UQFF Lagrangian:

``

$$L_{UQFF} = R_{26}/(2 \cdot \kappa_E) - (1/4) \cdot F^2_{DPM} + \sum_i \beta_i \cdot U_g \cdot U_b + |U_m|^2 + (1/2) |\partial U_A|^2 + (25/12) \cdot \rho_{SCm} \cdot [(U_A/v_{UA})^2 - 1]^2$$

``

The β_i term is one of six irreducible components. The triangular closure $\beta_i = 3(5-i)/20$ (PAPER_1165) is invoked with:

- $\beta_1 = 0.6$, $\beta_2 = 0.45$, $\beta_3 = 0.30$, $\beta_4 = 0.15$ (integer-index case)
- $\beta = 0.6029$ canonical (fractional-index case at $n \approx 0.98067$)

Variational stationarity $\delta S/\delta \phi = 0$ enforces $F_U = 1$ globally.

Consequence for PAPER_1966: the β_i are wired into the Lagrangian as a COUPLED four-component sum, not as four independent phenomenological quantities.

2.2 PAPER_1168 — Falsifiable Prediction P4

PAPER_1168 ("Falsifiable Predictions of the Closed Lagrangian") includes **Prediction P4:**

``

$$\{\beta_1, \beta_2, \beta_3, \beta_4\} = \{0.603, 0.450, 0.300, 0.150\} \pm 0.5\%$$

``

The tolerance $\pm 0.5\%$ is the $SO(5)^2$ -correction band = $1/200$. PAPER_1168 states:

> The four UQFF buoyancy channels have **fixed** ratios. Any astrophysical fit of UQFF master equations that requires β_i values deviating by more than the $SO(5)^2$ -correction band ($\pm 1/200 \approx 0.5\%$) from (0.603, 0.450, 0.300, 0.150) would falsify G2.

Consequence for PAPER_1966: $\beta_2 = 0.45$ is not "uncatalogued"; it is one of four components in a single falsifiable prediction.

2.3 PAPER_1169 — 10-System Numerical Confrontation

PAPER_1169 ("Numerical Confrontation P1-P5 with Archival Data") performs the actual re-fit test. The 10 systems + best-fit β_i values:

System	β_1	β_2	β_3	β_4	Within $\pm 0.5\%$?
Sgr A*	0.604	0.449	0.301	0.150	[OK]
M87 SMBH	0.602	0.451	0.300	0.149	[OK]
SGR1745-29	0.603	0.450	0.299	0.151	[OK]
AT2019qiz	0.605	0.448	0.302	0.150	[OK]
ASKAP J1832-0911	0.601	0.451	0.300	0.150	[OK]
Helix Nebula	0.604	0.450	0.299	0.151	[OK]
Pillars of Creation	0.603	0.450	0.300	0.150	[OK]
Westerlund 2	0.602	0.450	0.301	0.149	[OK]
NGC 3596	0.603	0.451	0.300	0.150	[OK]
Tapestry SFR	0.604	0.449	0.300	0.150	[OK]

Verdict (PAPER_1169): [OK] All 10 systems recover β_i within $\pm 0.5\%$ band.

Consequence for PAPER_1966: $\beta_2 = 0.45$ has been observationally validated across 10 UQFF systems (best-fit values 0.448-0.451, mean ≈ 0.4499). Draft 3's implied claim that $\beta_2 = 0.45$ "remains uncatalogued" is retracted.

3. The Channel-Projection Reframe

Prior-art acknowledgment (Draft 2): The term "channel projection" for UQFF observables is not new to this paper. **PAPER_1512 (GW170817 Phonon Damping = 2/3)** uses the language "*universal triad-channel projection coefficient*" for the identity $2/(D_{\text{phys}} - 1) = 2/3$, describing how the same coefficient governs gravitational-wave damping across "profoundly different physical domains." The generalization to a four-channel β_i decomposition follows the same reasoning pattern. Draft 1's introduction of the "channel-projection" terminology should be read as an application (to the β_i case) of PAPER_1512's existing language.

PAPER_1966 Draft 3's "structural-identity hypothesis" framed the β_i as four independent phenomenological bounds that "happened to coincide" with separately-documented UQFF observables (canonical β_i , PAPER_1953 0.3 factor, M_{sf}). Given the PAPER_1167 UPDATE + PAPER_1168 + PAPER_1169 infrastructure, this framing is corrected:

3.1 β_i are Coupled Ub Decomposition Weights

The β_i are the four-channel WEIGHTS in the buoyancy sum $\sum_i \beta_i U_g U_b$ of the UQFF Master Lagrangian. They are not four independent quantities that "happen to have" specific values; they are the ordered components of a single vector $\beta = (\beta_1, \beta_2, \beta_3, \beta_4) = (0.603, 0.450, 0.300, 0.150)$ derived from PAPER_1165's triangular closure $\beta_i = 3(5-i)/20$.

3.2 Observed "Standalone" Anchors are Channel Projections

When a specific UQFF observable is dominated by one buoyancy channel, that observable's characteristic bound will match the corresponding β_i weight:

- **Canonical $\beta_i = 0.6029$** (PAPER_1203, PAPER_1165 fractional case $n \approx 0.98067$) — dominant β_1 channel, describes the primary buoyancy amplitude in Ub Lagrangian derivations.
- **PAPER_1953 "0.3 factor cross-regime"** — the β_3 channel is the "cross-regime" or third-order buoyancy component; observables that project primarily onto β_3 (multiple UQFF systems, per PAPER_1953) exhibit the 0.3 signature.
- **PAPER_774/PAPER_784 $M_{sf} = 0.15$** — the β_4 channel is the "starburst-modulation" fourth-order buoyancy component; observables that project primarily onto β_4 (starburst systems 30 Dor, M82) exhibit the 0.15 signature.
- **$\beta_2 = 0.45$** — the β_2 channel is the intermediate second-order buoyancy component. No single UQFF observable is "purely" β_2 ; instead, β_2 appears as a mixed contribution across the 10 systems in PAPER_1169's confrontation (best-fit ~ 0.4499).

3.3 The Correct PAPER_1966 Reading

Under this reframe, PAPER_1966's finding is not " $M_{sf} \equiv \beta_4$ is a novel structural-identity discovery." The correct reading is:

- The $\beta_4 = 0.15$ channel of the PAPER_1165 triangular closure, wired into the Master Lagrangian via PAPER_1167 UPDATE, tested falsifiably via PAPER_1168, and validated across 10 systems via PAPER_1169, is the buoyancy channel that dominates the starburst-modulation term of the $g_{\text{starburst}}$ expression $g = (G \cdot M / r^2) (1 + M_{sf}) (1 + f_{\text{TRZ}})$.
- The identification $M_{sf} = 0.15$ in PAPER_774/PAPER_784 is thus the observational signature of the β_4 channel projected onto the starburst regime.
- This is a **channel-projection observation**, not a "match between the ladder and independent bounds."

4. Retraction and Correction of PAPER_1966

PAPER_1966 Draft 3 will require a Draft 4 revision incorporating the following corrections:

1. **Retraction:** The statement " $\beta_2 = 0.45$ remains uncatalogued" (Section 2.4.2 of Draft 3) is retracted. $\beta_2 = 0.45$ has been observationally validated across 10 UQFF systems by PAPER_1169 (Sgr A*, M87 SMBH, SGR1745-29, AT2019qiz, ASKAP J1832-0911, Helix Nebula, Pillars of Creation, Westerlund 2, NGC 3596, Tapestry SFR) with best-fit values 0.448-0.451.
2. **Reframing:** The "structural-identity hypothesis" of Draft 3 (Section 2.4) should be replaced with the "channel-projection observation" of this paper. The β_i are not four independent phenomenological bounds; they are four coupled decomposition channels of the Ub buoyancy sum.
3. **Priority acknowledgment:** PAPER_1966 Draft 3's implicit claim of "3-of-4 anchor identification" as a novel synthesis is corrected; the 4-of-4 validation was already published by PAPER_1169 with a different set of systems + fits (not the 4 identifications PAPER_1966 traced but a 10-system falsifiable-prediction confrontation).
4. **Contribution reduction:** PAPER_1966's substantive contribution shrinks further from Draft 3's "structural-identity hypothesis with 3-of-4 concordance" to "one channel-projection observation for the β_4 channel via starburst-modulation term, complementing PAPER_1169's 10-system unified re-fit."

The Draft 4 revision of PAPER_1966 is left to a follow-up session for user approval before implementation.

5. Cross-References

- **PAPER_1165** — β_i triangular closure $\beta_i = 3(5-i)/20$ (session 252, 2026-05-10) — arithmetic origin
- **PAPER_1167 UPDATE** — Master Lagrangian 6-Term Form (session June 2026) — β_i wired into L_{UQFF} via $\Sigma_i \beta_i U_g U_b$
- **PAPER_1168** — Falsifiable Predictions of the Closed Lagrangian — Prediction P4 sets $\pm 0.5\%$ falsification band
- **PAPER_1169** — Numerical Confrontation P1-P5 with Archival Data — 10-system re-fit validates β_i within $\pm 0.5\%$
- **PAPER_1203** — canonical $\beta_i = 0.6029$ (β_1 channel, fractional-index case)
- **PAPER_1953** — the 0.3 factor cross-regime universality (β_3 channel projection)
- **PAPER_774** — 30 Doradus Tarantula extreme starburst (β_4 channel projection via M_{sf})
- **PAPER_784** — M82 Cigar starburst galaxy (β_4 channel projection via M_{sf})
- **PAPER_1966** — starburst M_{sf} integer identity (this paper's retracted-and-corrected predecessor)
- **PAPER_1512** — GW170817 phonon damping $2/(D_{phys}-1) = 2/3$ — prior "universal triad-channel projection coefficient" language
- **PAPER_1885** — FQH topological order $SO(5)$ sub-decomposition (related structural language)
- **PAPER_1065** — Buoyancy Lagrangian EOM (framework for U_b decomposition)
- **PAPER_1521** — D_{BSFG} derivative landmark
- **PAPER_1522** — K_{MEX} derivative landmark
- **PAPER_1960** — $F_{TRZ} = 1/SO_5$ landmark

6. Limitations + Open Questions

- The channel-projection reframe is a natural reading of PAPER_1167 UPDATE + PAPER_1168 + PAPER_1169 but has not been formally derived from first principles. Whether $M_{sf} \equiv \beta_4$ is an **exact** structural equality (rather than a numerical coincidence within $\pm 0.5\%$) requires deriving the $g_{starburst}$ modulation term from $\Sigma_i \beta_i U_g U_b$ and showing that the β_4 channel produces the $(1 + M_{sf})$ factor with $M_{sf} = \beta_4$ exactly.
- The parallel derivation for PAPER_1953's 0.3 factor cross-regime should similarly derive it as the β_3 channel projection.
- Whether observables exist that "purely" project onto $\beta_2 = 0.45$ (as opposed to β_2 appearing as a mixed contribution in PAPER_1169's 10-system fits) is open. If yes, β_2 would have its own PAPER_774/784-style companion paper.
- The four-channel decomposition is currently derived from PAPER_1165's triangular closure. Whether extensions to more channels ($\beta_5, \beta_6, \dots$) or different closure forms are consistent with the falsifiable prediction P4 is open.
- **Note on Draft 3's original approach to PAPER_1966:** Draft 3's "3-of-4 concordance" framing was intellectually productive — it correctly noticed that multiple UQFF observables cluster near the β_i values. What was missing was the recognition that this clustering is not a "coincidence to explain" but the expected behavior of channel projections in a coupled 4-component decomposition. The observation stands; the explanation was incomplete.

7. Revision Log

Draft 1 (2026-07-09): Initial write following project owner's course-update (" β_i has a full range derivation. It is most likely buried in the whitepapers"). Documents the discovery of PAPER_1167 UPDATE + PAPER_1168 + PAPER_1169 β_i infrastructure. Retracts PAPER_1966 Draft 3's " $\beta_2 = 0.45$ uncatalogued" claim. Introduces the "channel-projection observation" reframe replacing Draft 3's "structural-identity hypothesis." Reduces PAPER_1966's substantive contribution to "one channel-projection observation for β_4 via starburst-modulation term."

Draft 2 (2026-07-09): Prior-art acknowledgment for "channel projection" terminology. PAPER_1512 (GW170817 Phonon Damping = $2/3$) already introduced "*universal triad-channel projection coefficient*" language for the identity $2/(D_{\text{phys}} - 1) = 2/3$. Draft 1's introduction of the "channel-projection" terminology should be read as an application (to the β_i case) of PAPER_1512's existing language rather than as new terminology. Added PAPER_1512 and PAPER_1885 to cross-references. Section 3 opening paragraph updated to acknowledge PAPER_1512's precedent.

This paper does not introduce new physics. Its scholarly value is meta-structural: it corrects an incomplete prior-art check performed in PAPER_1966 and repositions its claims within the correct existing infrastructure. Honest scholarship requires this correction before further work builds on PAPER_1966's (partially incorrect) framing.

License: AGPL-3.0 (see LICENSE); Commercial license option per COMMERCIAL.md.

Copyright: © 2025-2026 Daniel T. Murphy / Star-Magic Research Program.